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Code

JupyterLab Python 3 (ipykernel)

[7]:
```
# ==========================================
# Customer Segmentation using Unsupervised Learning
# Algorithms: K-Means, Hierarchical Clustering, DBSCAN
# ==========================================
%matplotlib inline
# 1. Import Required Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage

# 2. Load Dataset
data = pd.read_csv("Mall_Customers.csv")
print("First Five Rows of Dataset")
print(data.head())

# 3. Select Important Features
X = data[['Annual Income (k$)', 'Spending Score (1-100)']]

# 4. Feature Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# ==========================================
# 5. K-MEANS CLUSTERING
# ==========================================

# Find optimal number of clusters using Elbow Method
wcss = []
for i in range(1, 11):
```

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Code

JupyterLab Python 3 (ipykernel)

```python
# ==========================================

# Find optimal number of clusters using Elbow Method
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot Elbow Graph
plt.figure(figsize=(6,4))
plt.plot(range(1,11), wcss, marker='o')
plt.title("Elbow Method for Optimal K")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()

# Apply K-Means with K=5
kmeans = KMeans(n_clusters=5, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)

# Plot K-Means Clusters
plt.figure(figsize=(6,4))
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_kmeans)
plt.title("Customer Segmentation using K-Means")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.show()

# ==========================================
# 6. HIERARCHICAL CLUSTERING
# ==========================================

# Create Dendrogram
linked = linkage(X_scaled, method='ward')
```

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Code

JupyterLab Python 3 (ipykernel)

```python
# Create Dendrogram
linked = linkage(X_scaled, method='ward')
plt.figure(figsize=(8,5))
dendrogram(linked)
plt.title("Dendrogram for Hierarchical Clustering")
plt.xlabel("Customers")
plt.ylabel("Distance")
plt.show()

# Apply Agglomerative Clustering
hc = AgglomerativeClustering(n_clusters=5)
y_hc = hc.fit_predict(X_scaled)

# Plot Hierarchical Clusters
plt.figure(figsize=(6,4))
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_hc)
plt.title("Customer Segmentation using Hierarchical Clustering")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.show()

# ========================================
# 7. DBSCAN CLUSTERING
# ========================================

dbscan = DBSCAN(eps=0.5, min_samples=5)
y_db = dbscan.fit_predict(X_scaled)

# Plot DBSCAN Clusters
plt.figure(figsize=(6,4))
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_db)
plt.title("Customer Segmentation using DBSCAN")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
```

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Code

JupyterLab Python 3 (ipykernel)

```
# ==========================================

dbscan = DBSCAN(eps=0.5, min_samples=5)
y_db = dbscan.fit_predict(X_scaled)

# Plot DBSCAN Clusters
plt.figure(figsize=(6,4))
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_db)
plt.title("Customer Segmentation using DBSCAN")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.show()

# ==========================================
# 8. Print Cluster Labels
# ==========================================

print("\nK-Means Cluster Labels:")
print(y_kmeans)

print("\nHierarchical Cluster Labels:")
print(y_hc)

print("\nDBSCAN Cluster Labels:")
print(y_db)
```

```
First Five Rows of Dataset
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

Elbow Method for Optimal K

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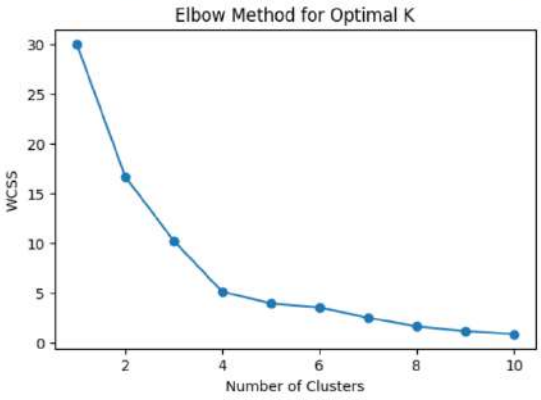
Code | JupyterLab | Python 3 (ipykernel)

```
print("\nHierarchical Cluster Labels:")
print(y_hc)

print("\nDBSCAN Cluster Labels:")
print(y_db)
```

```
First Five Rows of Dataset
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```



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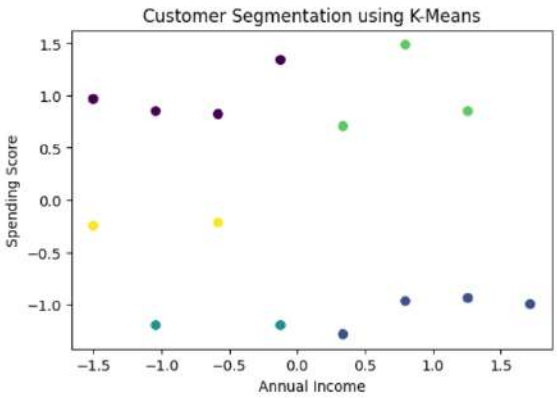
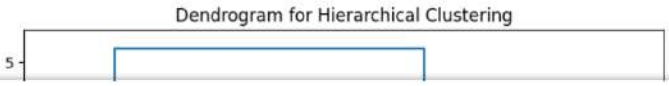
Code

JupyterLab Python 3 (ipykernel)

Number of Clusters





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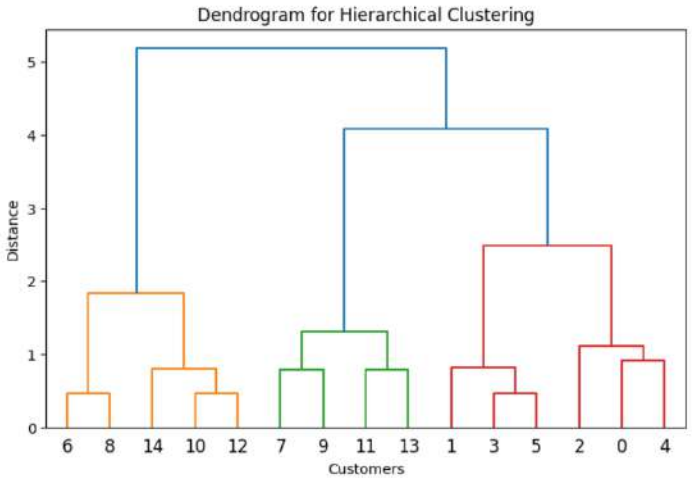
Code JupyterLab Python 3 (ipykernel)



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Code

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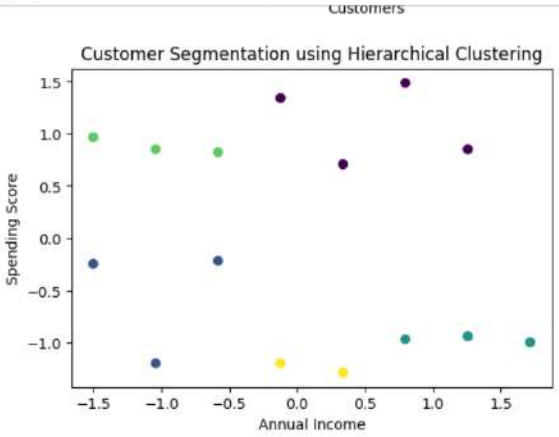

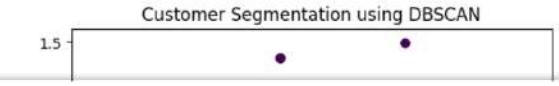

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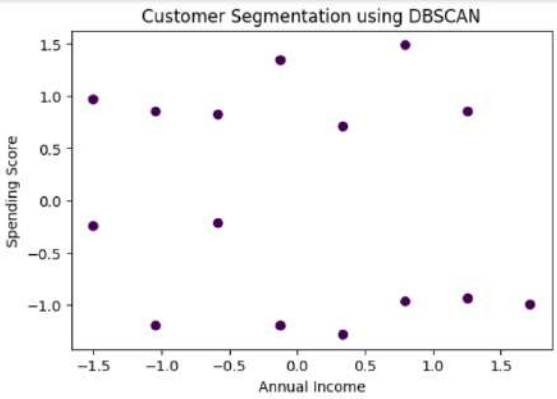
Code

JupyterLab Python 3 (ipykernel)

```
print("\nHierarchical Cluster Labels:")
print(y_hc)

print("\nDBSCAN Cluster Labels:")
print(y_db)
```




```
K-Means Cluster Labels:
[4 0 2 0 4 0 2 0 1 3 1 3 1 3 1]

Hierarchical Cluster Labels:
[1 3 1 3 1 3 4 0 4 0 2 0 2 0 2]
```

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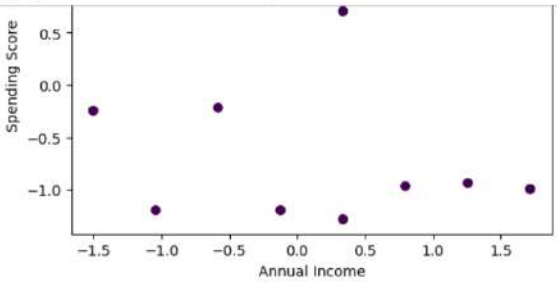
Code JupyterLab Python 3 (ipykernel)




```
K-Means Cluster Labels:
[4 0 2 0 4 0 2 0 1 3 1 3 1 3 1]

Hierarchical Cluster Labels:
[1 3 1 3 1 3 4 0 4 0 2 0 2 0 2]

DBSCAN Cluster Labels:
[-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1]
```

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